

UDITYA NARAYAN TIWARI

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Uttar Pradesh, India

Summary

B.Tech student specializing in **Artificial Intelligence and Machine Learning** with hands-on experience in building and deploying **end-to-end AI/ML applications**. Proficient in **Python, Deep Learning, FastAPI, Docker**, and cloud platforms including **AWS**. Experienced in developing scalable solutions, **REST APIs**, and data-driven applications with **Generative AI** while following industry best practices. Strong foundation in **Data Structures and Algorithms** with proven problem-solving ability through competitive programming, complemented by effective communication, collaboration, and continuous learning.

Technical Skills

Languages: Python, C++, Java, SQL

AI & Machine Learning: Machine Learning, Deep Learning, **NLP**, **Feature Engineering**, Model Evaluation, Predictive Analytics

Generative AI: LLMs, **RAG**, **Embeddings**, Semantic Search, **Vector Databases**

Frameworks & Libraries: LangChain, Scikit-learn, TensorFlow, Transformers, Sentence Transformers, FastAPI, Flask, Streamlit

Data Science: Pandas, NumPy, Matplotlib, **FAISS**, **ChromaDB**

Cloud & Infrastructure: Amazon Web Services (AWS), Docker

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Kaggle

Projects

NeuroLens: Knowledge Retrieval Engine — *Python, LangChain, RAG, LLMs, FAISS, ChromaDB*

Live

- Developed an AI-powered Retrieval-Augmented Generation (RAG) platform enabling **context-aware question answering** across PDFs, documents, and web content using Large Language Models (LLMs).
- Engineered semantic search pipelines utilizing vector embeddings with FAISS and ChromaDB to deliver **fast, accurate, and scalable knowledge retrieval**.
- Designed **automated document ingestion, indexing, and retrieval workflows**, significantly improving information discovery through intelligent context-aware responses.

Smart Medical Care for Rural Areas — *Python, NLP, Flask, Sentence Transformers*

Live

- Developed an AI-powered medicine recommendation system that provides **symptom-based medicine recommendations** using Natural Language Processing (NLP).
- Implemented **cosine similarity** and **Maximal Marginal Relevance (MMR)** ranking to improve semantic matching accuracy between symptoms and medicines.
- Built robust **data preprocessing pipelines** and deployed the application using Flask, enabling scalable real-time recommendation through RESTful APIs.

Breast Cancer Prediction using Machine Learning — *Python, Scikit-learn, Jupyter Notebook*

Live

- Developed a supervised machine learning pipeline for breast cancer diagnosis using the **UCI Breast Cancer Wisconsin dataset** containing **569 samples and 30 clinical features**.
- Implemented and benchmarked **Logistic Regression, Decision Tree, Random Forest, SVM, and k-NN** models using Scikit-learn.
- Achieved approximately **97% classification accuracy** while evaluating model performance using **ROC-AUC, confusion matrix, precision, recall, and F1-score**.

Experience

Microsoft Learn Student Chapter
Technical Team Member & Community Admin

VIT Bhopal University, Sehore, MP
04/2025 – 04/2026

- Planned and executed **technical workshops, coding competitions, and hackathons**, promoting hands-on learning in Microsoft technologies, cloud computing, and AI.
- Led **community engagement initiatives** by coordinating with student developers, organizing technical sessions, and fostering knowledge sharing across the campus community.

Matrix Media Club
Core Member, Tech Team

VIT Bhopal University, Sehore, MP
01/2025 – 07/2025

- Collaborated with **cross-functional teams** to design and implement technology-driven solutions that **improved communication workflows** between organizers and stakeholders.
- Contributed to the **development and deployment of technical solutions** while supporting the planning, execution, and successful delivery of club events and initiatives.

Education

VIT Bhopal University
B.Tech - CSE (AI & ML) | **CGPA: 9.00 / 10.00**

Sehore, Madhya Pradesh
2023 – Present

Achievements & Certifications

- Microsoft Azure Data Fundamentals:** Cloud Computing and Business Development
- Indian Institute of Remote Sensing (IIRS), ISRO:** AI/ML for Geodata Analysis
- International Hackathon:** Final Round Qualifier at the VIT Bhopal × Johns Hopkins University International Hackathon
- Hackathon Achievement:** First Prize Winner at the VIT Bhopal University Hackathon